Pre-analysis Plan

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Pre-analysis Plan – Hypothesis 1

Hypothesis 1 posits that men who have daughters express stronger support for gender equality in employment and caregiving roles.

To test this hypothesis, the dependent variable is **gender_equality_support**, constructed from responses to attitudinal survey items in two waves. In 2018, the item (*plh0308_v2* of pl.dta) was measured on a 7-point scale ranging from strong disagreement to strong agreement with the statement: "Ideally, men and women should be equally employed so that both can take care of the household and family." These values will be used without transformation. In 2012, a conceptually similar item (*plh0308_v1* of pl.dta) was asked using a 4-point scale. To ensure comparability across years, the 2012 responses will be reverse-coded and linearly rescaled onto a 7-point scale, so that higher values uniformly indicate greater support for gender equality. Respondents with missing or non-substantive responses (e.g., "not applicable," "not asked," or invalid codes) will be excluded from the analysis.

The key independent variable is a binary indicator for whether the respondent has at least one daughter (**Dau_Dummy2**). This variable is derived from the variable *pld0025* of pl.dta, which asks whether the respondent has any daughters. The original variable is only available for select survey years (2001, 2006, 2011, 2016), so to extend coverage to intervening years, daughter status is imputed for the years in between by carrying forward values from the most recent available wave.

To control for the presence of sons, we construct an analogous binary variable **Son_Dummy2**, based on the *pld0027* item, which asks whether the respondent has any sons. This variable is constructed and imputed in the same way as the daughter variable.

Two additional covariates capture the respondent's parental background. **IsDadGerman** is

coded as 1 if the respondent's father is of German nationality (*fnat = 1* of bioparen.dta) and 0 otherwise. Similarly, **IsMomGerman** is coded as 1 if the respondent's mother is German (*mnat = 1* of bioparen.dta).

These variables help account for cultural influences on attitudes toward gender roles.
The analysis is restricted to male respondents (*sex = 1* of biobirth.dta), as the hypothesis specifically concerns whether men's attitudes toward gender equality are influenced by their experience of raising daughters. Observations with missing or invalid values on any key variables are excluded.

The analysis will estimate an OLS regression of gender_equality_support on Dau_Dummy2, Son_Dummy2, and the parental nationality controls. The model will use standard errors clustered by individual ID (*pid*) to adjust for possible intra-individual correlations. The key test of the hypothesis is the coefficient on Dau_Dummy2. A positive and statistically significant coefficient ($p < .05$) would support the hypothesis. A null or negative coefficient would be interpreted as not supporting the hypothesis.

Pre-analysis Plan – Hypothesis 2

Hypothesis 2 posits that having daughters leads to more left-leaning political preferences. This hypothesis is tested by examining whether men with daughters report a more left-leaning position on the political spectrum than those without daughters.

The outcome variable (**Y**) is **Support_Left**, constructed by reversing the original 11-point political ideology scale (*plh0004* of pl.dta), where 0 indicates the most left-leaning and 10 the most right-leaning political orientation. The transformation (*10 - plh0004*) makes higher values indicate stronger left-leaning attitudes, aligning the direction of the scale with the hypothesis.
Observations are excluded if *plh0004* has negative values, which represent missing responses or design-based non-inclusions. The analysis is restricted to male respondents (*sex = 1* of biobirth.dta), and only those with valid data for daughter and son status are included.

The primary independent variables are the same as in Hypothesis 1. The main variable of interest is **Dau_Dummy2**, coded as 1 if the respondent has at least one daughter and 0 otherwise. Control variables include **Son_Dummy2** (a binary indicator for having at least one son), and two indicators for parental nationality—**IsDadGerman** and **IsMomGerman**—to adjust for cultural and political socialization effects stemming from family background.

The hypothesis is tested using a linear regression model with standard errors clustered by individual identifier (*pid*). The key statistic is the coefficient on *Dau_Dummy2*. A positive and statistically significant estimate supports the hypothesis that having daughters is associated with more left-leaning political views. A non-significant or negative coefficient would suggest no such effect or the opposite.

Pre-analysis Plan – Hypothesis 3

Hypothesis 3 states that having daughters leads to greater gender equality in the household. This is operationalized by testing whether men with daughters spend more time on housework over the weekend—a behavioral indicator of a more equitable division of domestic labor. Specifically, we examine whether weekend housework time (Z) is greater for men with daughters than for those without.

The outcome variable is **weekend**, created by summing two harmonized variables: *pli0012_h* (hours spent on housework on Saturday) and *pli0016_h* (hours on Sunday) of pl.dta. Only values greater than or equal to zero are retained, excluding implausible or missing responses. These values are stored in intermediate variables *h1* and *h2*, and their sum becomes *weekend*, capturing total weekend housework time.

The primary independent variables are the same as in Hypothesis 1. The main variable of interest is **Dau_Dummy2**, coded as 1 if the respondent has at least one daughter and 0 otherwise. Control variables include **Son_Dummy2** (a binary indicator for having at least one son), and two indicators for parental nationality—**IsDadGerman** and **IsMomGerman**—to adjust for cultural and political socialization effects stemming from family background.

The analysis is restricted to male respondents (*sex = 1* of biobirth.dta) to isolate the effect of daughters on men's household behavior. The sample includes only those with valid responses for daughter and son status and valid weekend time values (≥ 0).

The hypothesis will be tested using a linear regression model with cluster-robust standard errors at the individual level (clustered by *pid*). The coefficient on Dau_Dummy2 is the key estimate. If it is positive and statistically significant, it will be interpreted as supporting the hypothesis that having daughters increases men's contribution to housework, indicating greater gender equality in household practices. If the coefficient is not statistically different from zero or is negative, it will be considered evidence against the hypothesis.

Pre-analysis Plan – Hypothesis 4

Hypothesis 4 investigates whether the effect of having daughters on support for gender equality in the household is stronger for male respondents than for female respondents. This is a moderation hypothesis: the gender of the respondent moderates the relationship between having daughters and support for gender equality.

The dependent variable (**X**) is **weekend**, a continuous variable representing the total number of hours a respondent reports doing housework over the weekend. It is created by summing *pli0012_h* (Saturday housework hours) and *pli0016_h* (Sunday housework hours) of pl.dta, both of which are restricted to non-negative values. Missing or implausible values are excluded by conditioning on $weekend \geq 0$.

Some independent variables are common to Hypothesis 1. The main variable of interest is **Dau_Dummy2**, coded as 1 if the respondent has at least one daughter and 0 otherwise. Control variables include **Son_Dummy2** (a binary indicator for having at least one son) and two indicators for parental nationality—**IsDadGerman** and **IsMomGerman**—to adjust for cultural and political socialization effects stemming from family background. In addition, this model includes the moderator variable **Male**, coded as 1 for male respondents and 0 for female respondents. An interaction term between **Dau_Dummy2** and **Male** is included to test whether the effect of having daughters differs by respondent gender.

The regression model is estimated using OLS with standard errors clustered by individual ID (*pid*). The central test is the interaction term between Dau_Dummy2 and Male. A positive and statistically significant coefficient on the interaction term would support the hypothesis that men show a stronger increase in gender-equal behavior (measured by housework time) after having daughters, compared to women. If the interaction term is insignificant or negative, the hypothesis would not be supported.